TradeIO

Requirements Document

Trading Journal and Behavioral Analytics Platform

Senior Design

Assignment #9: Requirements Document and Technical Specification Document

Team Members

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Project Goal

TradeIO is a trading journal intended to support structured trade recording, evaluation of execution and psychology, risk awareness, and evidence-based behavioral review.

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1. Version and Change Log

Version	Date	Change	Authors
0.1	October 30, 2026	Team-approved requirements and acceptance thresholds based on user stories, D0–D2, and project planning.	Ishaa Jain; Samarth Prajapati

2. Purpose and Scope

TradeIO is proposed as a web-based trading journal and behavioral analytics platform. Its purpose is to help traders document transactions, compare outcomes with their intended trading process, evaluate adherence to personal risk controls, and review recurring behavioral patterns. Financial results will provide context for decisions; they will not serve as the sole measure of trade quality.

The initial scope prioritizes account-specific trade records, manual entry, accessible history, deterministic performance calculations, and regional settings. Journaling, risk controls, accountability, coaching, and export capabilities are categorized according to their proposed release priority. Direct order placement, investment recommendations, and brokerage account control fall outside the intended scope. Optional broker-data integration remains subject to a separate feasibility decision.

3. Stakeholders and User Goals

The primary stakeholders are individual traders who require reliable records and interpretable feedback. Samarth Prajapati represents a USD-based trading workflow; the two team members' fathers represent INR-based workflows in India. Their needs establish that account currency, market conventions, and local reporting dates cannot be assumed to be uniform.

Accountability partners are secondary stakeholders who may receive limited notices and, subject to prior trader authorization, initiate temporary restrictions within TradeIO. The development and administration team comprises Ishaa Jain and Samarth Prajapati. Additional stakeholder considerations include privacy-sensitive users, users of assistive technologies, and traders whose brokerage platform or instrument is not initially supported.

The Week 4 stories define five user goals: accurate trading records (US-01), understanding behavior (US-02), voluntary accountability (US-03), optional access to supported broker data (US-04), and currency/time-zone flexibility (US-05). Broker authorization and accountability cooldowns are further described in UC-01 and UC-02.

4. Glossary

Term or role	Definition
Trader	Authenticated user who owns and manages trading information.
Trading account	User-owned account record containing currency, balance, trades, and configuration.
Accountability partner	Person explicitly authorized to receive selected alerts or request a limited TradeIO cooldown.

Term or role	Definition
System administrator	Member of the TradeIO maintenance team with restricted administrative responsibilities.
Broker	Independent service through which actual market orders are executed.
Trade	Recorded trading position or transaction.
Fill	Individual execution contributing to an entry or exit.
Realized P&L	Net profit or loss attributed to closed trading activity.
Drawdown	Reduction from a prior equity peak under a defined calculation basis.
R-multiple	Outcome relative to the initial amount at risk, when that quantity is known.
Trade cap	Trader-defined daily maximum number of trades.
Cooldown	Time-limited restriction affecting specified TradeIO operations, not external broker activity.
Behavioral summary	Deterministically calculated account-level description of trading and psychology records.
AI insight	Optional explanatory text derived from predefined summary data.
IANA time zone	Recognized regional time-zone identifier used for local reporting boundaries.
ISO 4217	Standard three-letter currency-code system.
MoSCoW	Priority classification: Must, Should, Could, and Won't for this release.

5. Functional Requirements

Each requirement states an expected and verifiable behavior. The D2 references identify the earlier provisional requirement lineage; they do not indicate implementation completion.

ID	System requirement	Priority	Source	Verification
REQ-01	The system shall verify an authenticated session for each protected request and enforce ownership of the requested account. Missing sessions shall return HTTP 401; requests for another user's account shall return HTTP 403.	Must	US-01, US-04	Authorization test
REQ-02	The system shall permit an authenticated trader to register one or more trading accounts, each with an account currency, starting balance, and separately maintained trade history.	Must	US-01, US-05	Account creation test
REQ-03	The system shall accept a manually entered trade containing an owned account, instrument, direction, time, positive entry price, and positive quantity. Invalid submissions shall not create a trade record.	Must	US-01; UC-01	Validation test
REQ-04	The system shall retrieve trades belonging to the selected account, support an optional date range, and return pages of 1 to 100 records without exposing another account's records.	Must	US-01	Filter and access test
REQ-05	The system shall permit dated reflections associated with a trade or recorded independently. Unprovided psychology scores shall remain null rather than be inferred.	Should	US-01, US-02	Journal data test
REQ-06	The system shall calculate realized net profit or loss, closed-trade count, and win rate for a selected account and period using stored transactions. The win rate shall be reported as not applicable when no trades are closed.	Must	US-01, US-02	Calculation fixtures
REQ-07	The system shall validate account risk percentages within the inclusive range 0 to 100 and require any configured daily trade cap to be a positive integer. An invalid update shall preserve the previous configuration.	Should	US-02, US-03	Settings boundaries

ID	System requirement	Priority	Source	Verification
REQ-08	The system shall evaluate configured daily and weekly thresholds against periods defined by the trader's selected time zone. A value equal to a limit shall be identified as reaching that limit; open-position values shall not enter realized profit or loss.	Should	US-02, US-03	Boundary fixtures
REQ-09	The system shall present recent account trades and available deterministic performance results. When a result is absent or outdated, the interface shall distinguish that condition from current calculated data.	Must	US-01, US-02	Dashboard states
REQ-10	The system shall permit weekly and monthly goals to be defined with a target and period, and shall calculate progress from the relevant account records.	Should	US-01, US-02	Goal lifecycle test
REQ-11	When optional AI coaching is enabled, the system shall request a periodic narrative based on a deterministic behavioral summary. A failed AI request shall not prevent access to the underlying summary.	Could	US-02	Service outage test
REQ-12	The system shall register a trade-limit exceedance once per event and issue no more than one notification to an eligible configured accountability partner. Without such a partner, only the trader-facing warning shall apply.	Should	US-03; UC-02	Alert deduplication
REQ-13	The system shall present monetary records in the account's configured ISO 4217 currency, including USD and INR, and shall not combine different currencies into a single total without an approved conversion rule.	Must	US-05	Currency test
REQ-14	The system shall validate IANA time-zone identifiers, preserve event timestamps as UTC instants, and determine local reporting-day boundaries using the trader's configured time zone.	Must	US-05	Time-zone fixtures
REQ-15	If a supported broker integration is authorized for the release, the system shall import each external trade at most once per provider trade identifier and preserve manual entry when the provider is unavailable.	Could	US-04; UC-01	Import simulation
REQ-16	The system shall accept an accountability cooldown only from a previously authorized partner, for a duration from 10 minutes through 24 hours, and shall end the TradeIO restriction when the recorded expiration is reached.	Should	US-03; UC-02	Permission and expiry
REQ-17	The system shall record exactly one cap-reached event when the configured daily trade count reaches its threshold, present an in-application warning, and initiate no more than one notification to an authorized accountability partner for that event.	Should	US-03	Cap policy test
REQ-18	If multi-fill support is adopted for the release, the system shall retain the quantity, price, fee, and execution time of each individual entry and exit fill.	Could	US-01	Fill reconciliation
REQ-19	The system shall provide an account-authorized export of trading history or journal information in a documented file format, with no records from other users included.	Could	US-01, US-02	Export access test

6. Non-Functional Requirements

The team has approved the following numerical acceptance thresholds for the planned system. Compliance will be evaluated during future verification; no results are claimed at this stage.

ID	System requirement	Priority	Source	Verification
NFR-01	The system shall return the first 50 trade-history records within 2 seconds at the 95th percentile under 20 concurrent users against a test account containing 2,500 trades.	Should	US-01	Load test
NFR-02	The system shall deny all 30 scripted unauthorized requests across account, trade, journal, and settings resources without disclosing any protected record.	Must	US-01, US-04	Security suite
NFR-03	At least four of five first-time participants shall complete the prescribed manual trade-entry task within 3 minutes, using the same task instructions.	Should	US-01	Usability study

ID	System requirement	Priority	Source	Verification
NFR-04	For 100 requests bearing the same idempotency key, including 10 concurrent requests, the system shall retain exactly one trade record.	Must	US-01	Concurrent replay
NFR-05	Trade entry and deterministic analytics shall remain operable in all 10 controlled AI-provider outage scenarios.	Must	US-02	Fault injection
NFR-06	All 25 specified monetary test cases shall reproduce the expected exact-decimal results without binary floating-point discrepancies.	Must	US-01	Decimal fixtures
NFR-07	All three scripted sign-in, trade-entry, and history workflows shall be operable by keyboard alone, with labeled fields and error feedback independent of color.	Should	US-01	Accessibility audit
NFR-08	None of 20 inspected outbound AI payloads, notifications, and application error logs shall contain credentials, full journal narratives, or private account identifiers unnecessary for that operation.	Must	US-02, US-04	Payload inspection
NFR-09	All 10 simulated failures of optional broker or notification services shall leave committed trade records unchanged and present an appropriate failure or fallback state.	Should	US-03, US-04	Dependency tests

7. MoSCoW Prioritization

Must. REQ-01–04, REQ-06, REQ-09, REQ-13–14; NFR-02, NFR-04–06, NFR-08. These establish protected access, accurate trade records, core analysis, and correct regional handling.

Should. REQ-05, REQ-07–08, REQ-10, REQ-12, REQ-16–17; NFR-01, NFR-03, NFR-07, NFR-09. These extend the core journal with structured reflection, risk management, accountability, and quality controls.

Could. REQ-11, REQ-15, REQ-18–19. These remain conditional on time, provider access, and release-scope decisions.

Won't for this release. Direct order execution, trading signals, profit guarantees, public access to private journals or screenshots, unapproved cross-currency aggregation, native mobile applications, and unrestricted accountability-partner access are excluded.

8. Inherited Design Constraints

ID	Category	Constraint	Implication
CON-01	Economic	The project shall use student-accessible services and limit AI usage and storage expenditure.	Favors a web-first application and manual trade entry over expensive native and paid-data services.
CON-02	Ethical	Insight generation shall remain advisory and evidence-based, with suitable limitations disclosed.	Excludes market signals, guaranteed profits, automated orders, and unsupported behavioral claims.
CON-03	Legal and privacy	Trading records, reflections, and screenshots shall remain private and isolated by account ownership. The original MVP scope excluded direct broker connectivity.	Requires protected access and a separate decision before any optional broker integration enters scope.

ID	Category	Constraint	Implication
CON-04	Security	Protected access shall require verified identity, server authorization, and restricted file storage.	Excludes unsecured account access and publicly exposed private uploads.

9. Assumptions and Open Questions

The team has approved the requirements, MoSCoW priorities, and non-functional acceptance thresholds in this version. The following implementation policies and interface details remain open; the dates are planning targets.

Subject	Owner	Target date	Decision required
Broker integration	Samarth; advisor	Nov. 4, 2026	Decide whether a single supported provider is feasible as a stretch goal; preserve manual entry.
Multiple fills	Samarth	Nov. 4, 2026	Confirm whether individual fills require a child table in the first release.
Accountability policy	Ishaa	Nov. 4, 2026	Define which TradeIO actions a cooldown restricts; establish whether additional trade logging after the daily cap is prohibited or permitted with a recorded override; and confirm the settings-change lock. The decision will govern a subsequent requirement revision.
Numerical quality targets	Ishaa; Samarth	Oct. 30, 2026	Approved by both team members for use as the Week 10 acceptance baseline; validation is pending.
Additional API contracts	Ishaa; Samarth	Nov. 4, 2026	Resolve goal modification, notification, and optional broker import contracts.
Advisor review	Ishaa; Samarth	Upcoming weeks	The team plans to meet with Dr. William Hawkins III in the coming weeks. The date, comments, and resulting revisions will be recorded after the meeting.

10. Controlled References

User Stories and Use Cases (Week 4); D0 High-Level System Design, D1 Detailed Design, and D2 Detailed Design (Design Diagrams (D0, D1, D2)); Implementation Planning (Week 8); Technical Specification, version 0.1.